# UEE2718 – Project Phase - 1

## Project Review - 3

# AI Data Analytics Layer for Surveillance Systems

## Using Edge AI for Energy Management Systems

*Edge-AI occupancy intelligence for building energy management*

**Batch No: B2**


*Presented by*

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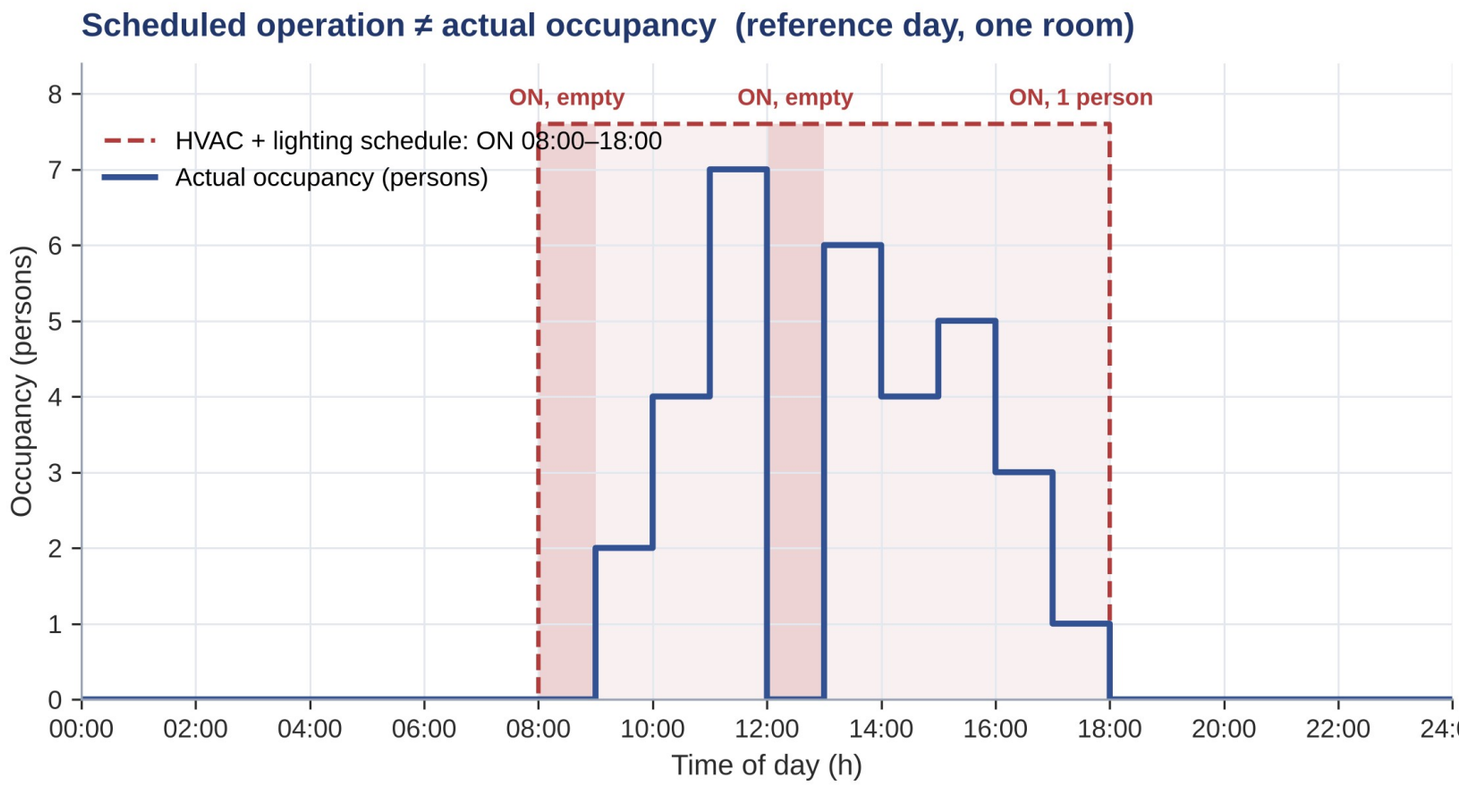
**Project Guide (Internal): Dr. Rajesh Panda, Prof./EEE, SSNCE**

# Objectives of the Project

| | | |
|---|---|---|
| 01 | Detect people live on a Jetson Nano edge node | IMPLEMENTED |
| 02 | Convert detections into a stable occupancy signal | IMPLEMENTED |
| 03 | Map occupancy to HVAC and lighting states | MODELLED |
| 04 | Quantify energy, cost and $CO_2$ with one simulation model | MODELLED |
| 05 | Test robustness and sensitivity of the savings | MODELLED |
| 06 | Define the metering and BAS/BMS path | FUTURE |

*Sense on the edge, decide with occupancy, prove the impact by simulation, then by metering.*

# Problem Formulation & Justification




*ILLUSTRATIVE REFERENCE DAY · model v1*

**82 %** power cut when a scheduled room is empty (1.70 → 0.30 kW, model)

**76 %** of business hours, smart meeting rooms sat unused (Schneider Electric, London, 2024 data)

**24 °C** BEE default AC set-point since 2020 (our Comfort set-point)

***A fixed schedule cools and lights the room whether 0 or 7 people are inside.***

# Literature Review & Research Gap

## Vision sensing

Hu et al. 2023 (IEEE IoT J.) · Tien et al. 2021 (Energies)

CCTV + deep learning can detect occupants in real buildings.

## Occupant counting

Choi et al. 2022 · Calautit et al. 2024

Counts, not just presence, drive HVAC demand.

## Control experiment

Kong et al. 2022 (Applied Energy)

Side-by-side test: **17–24 %** weekly HVAC savings, >80 % comfort satisfaction.

## Field study

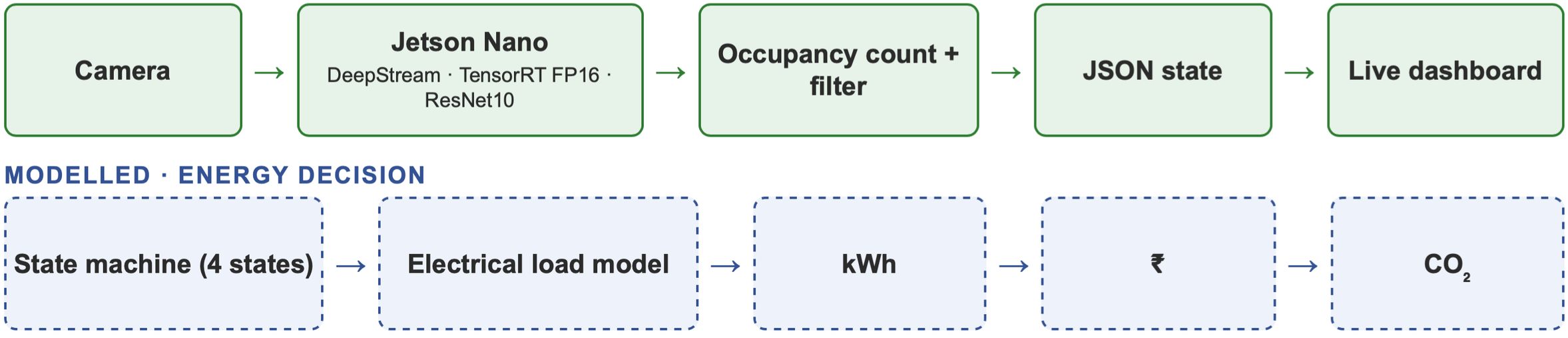
Lin et al. 2023 (Energies, LBNL)

AHU run-time cut 2 h 35 min/day; **6.1 % projected** annual whole-building saving (calibrated model).

***Prior work proves sensing and control separately. We build a reusable edge layer: existing camera → occupancy → quantified energy decision.***

# Block Diagram of the Project

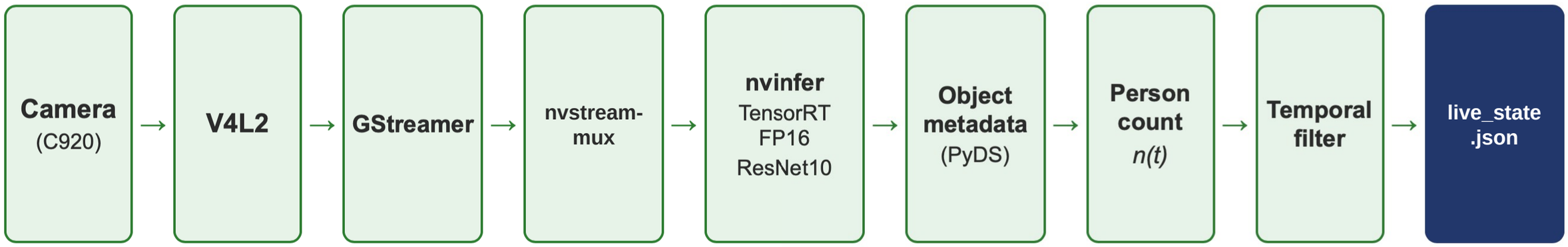
IMPLEMENTED · MODELLED · FUTURE




***The AI does not save electricity. It produces occupancy intelligence; the energy decision saves electricity.***

IMPLEMENTED

# Methodology / Work Flow: Inside the Jetson




## Edge

Video stays on the device; only a count leaves.

## FP16 TensorRT

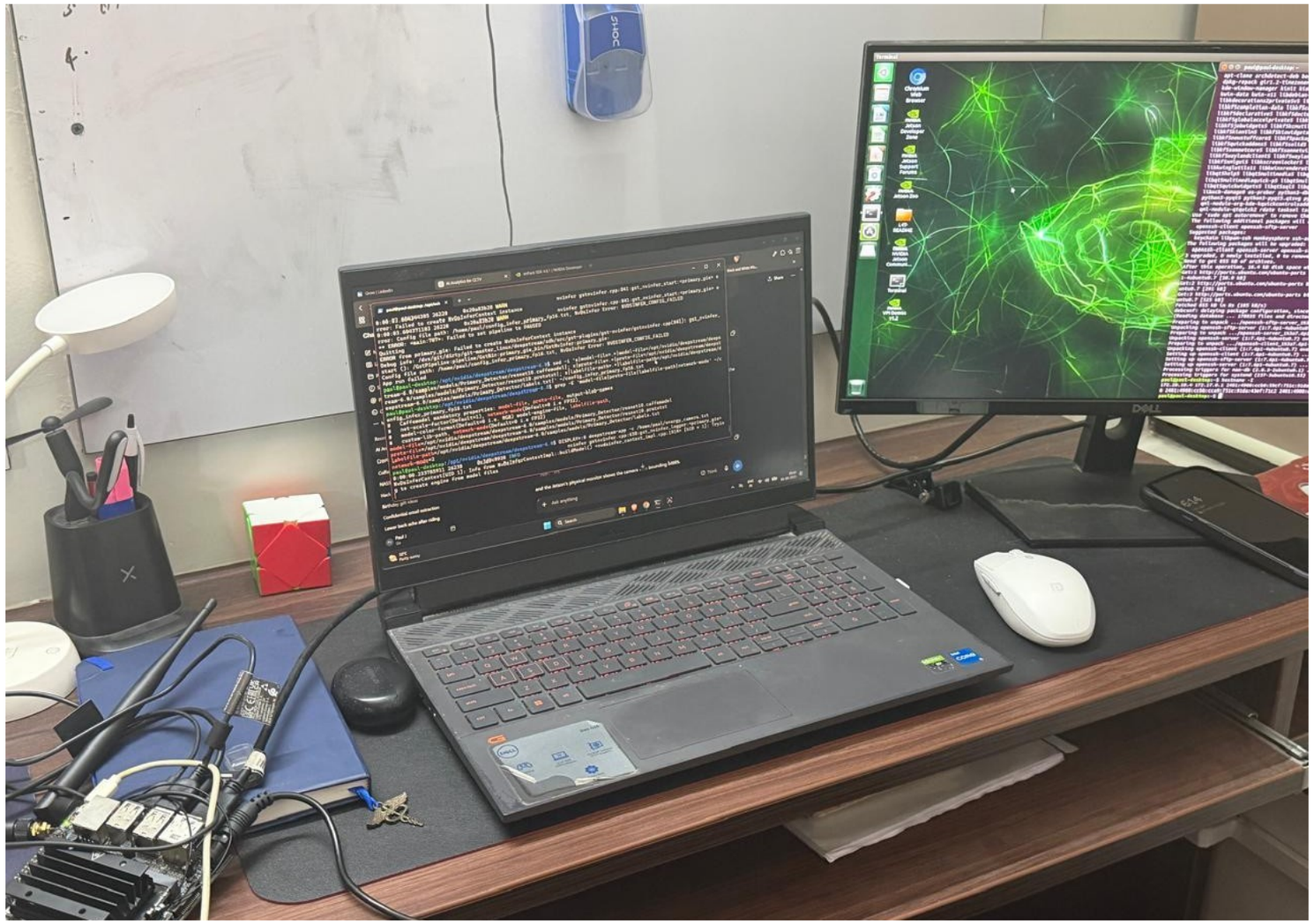
Fast inference on the Nano's 128-core GPU.

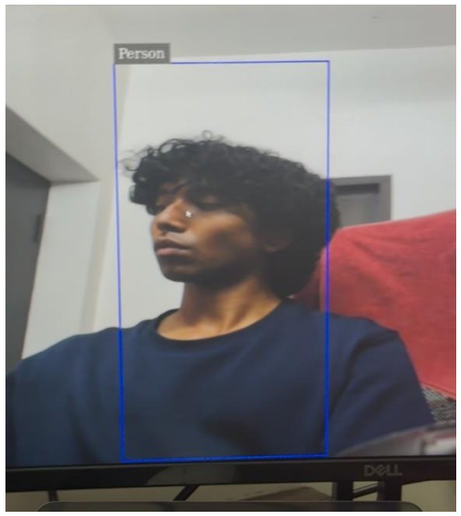
## ResNet10

Lightweight NVIDIA reference person detector.

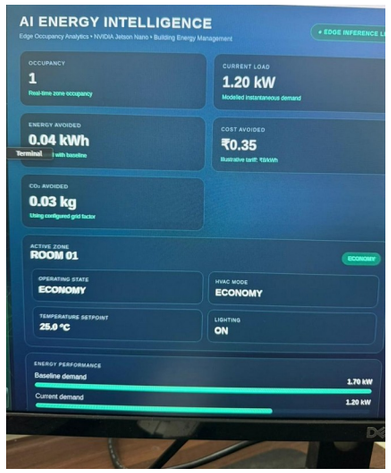
***Only metadata leaves the pipeline: a person count every second.***

# Prototype & Live Output






*Live detection*

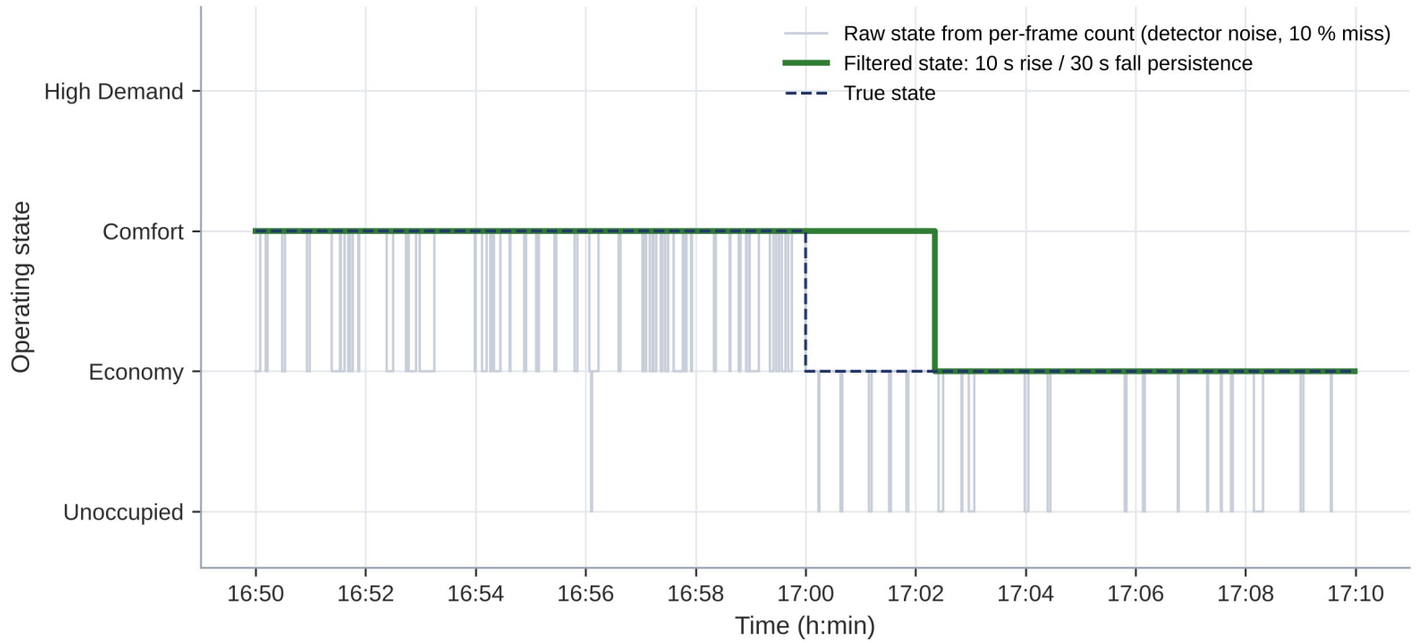



*Jetson Nano 4 GB · Logitech C920 · DeepStream 6.0.1 · TensorRT FP16 · ResNet10 · node cost ₹17,500*

***Detection, occupancy, state and modelled load update live on the Jetson.***

# Occupancy Intelligence: Temporal Filter

**Why temporal filtering: raw detections chatter, filtered state is stable**



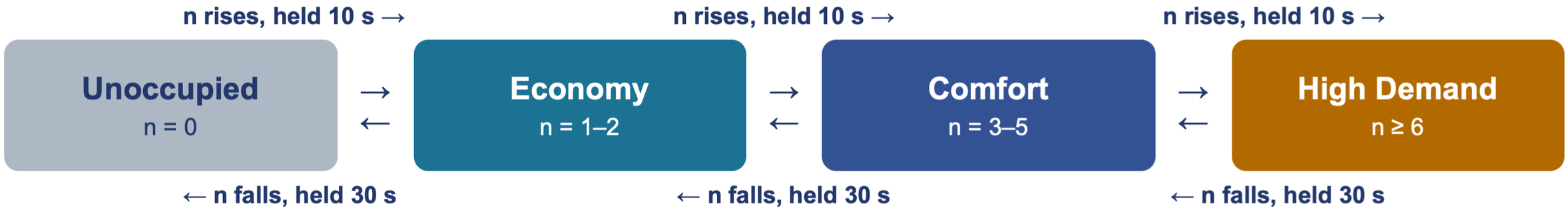

*SIMULATED DETECTOR NOISE (assumed 10 % per-person miss, not measured) · model v1*

| Filter | State changes / day | Time in correct state |
|---|---|---|
| No filter | 2,633 | 89.0 % |
| **Ours: 10 s rise / 30 s fall** | **9** | **99.1 %** |
| True value | 7 | 100 % |

*Simulated, 10 % detector miss assumed.*

***Our 10 s / 30 s filter removes 99.7 % of state flicker: fast to serve people, slow to switch off.***

# Occupancy → Energy State Machine




| Occupancy | State | Set-point | Lighting | Power |
|---|---|---|---|---|
| 0 | Unoccupied | 27.0 °C | OFF | **0.30 kW** |
| 1–2 | Economy | 25.0 °C | ON | **1.20 kW** |
| 3–5 | Comfort | 24.0 °C | ON | **1.70 kW** |
| 6+ | High Demand | 23.5 °C | ON | **2.00 kW** |

*Power values are modelling assumptions for one room (HVAC + 10 × 20 W lights).*

***This is where occupancy becomes an electrical quantity: n(t) → state → kW.***

# Electrical Energy Model

1 $P(t) = P_{HVAC}(t) + P_{Light}(t)$

2 $E = \Sigma\, P(t) \cdot \Delta t$

3 $E_{saved} = E_{base} - E_{ctrl}$

4 $Cost = E_{saved} \times Tariff$

5 $CO_2 = E_{saved} \times EF$

**WORKED EXAMPLE · REFERENCE DAY**

$E_{base}$ = 1.70 kW × 10 h

**17.0 kWh**

$E_{ctrl}$ = 0.30×2 + 1.20×2 + 1.70×4 + 2.00×2

**13.8 kWh**

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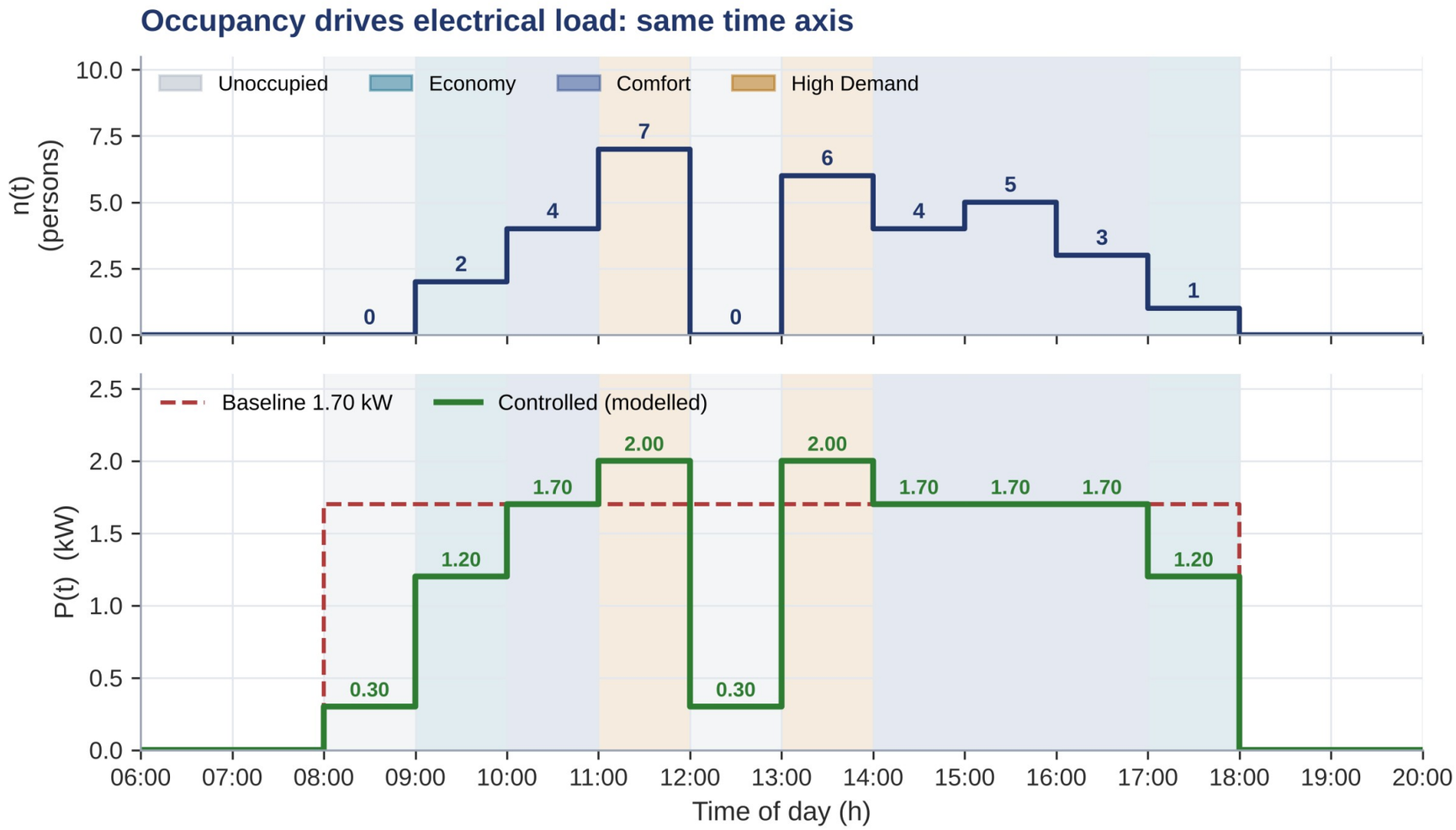
$E_{saved}$

**3.2 kWh/day (18.8 %)**

*Tariff ₹8/kWh (assumed) · EF 0.710 kg $CO_2$/kWh (CEA, FY 2024-25) · 10 h/day · 365 days*

***Every number in this deck comes from these equations and one simulation dataset.***

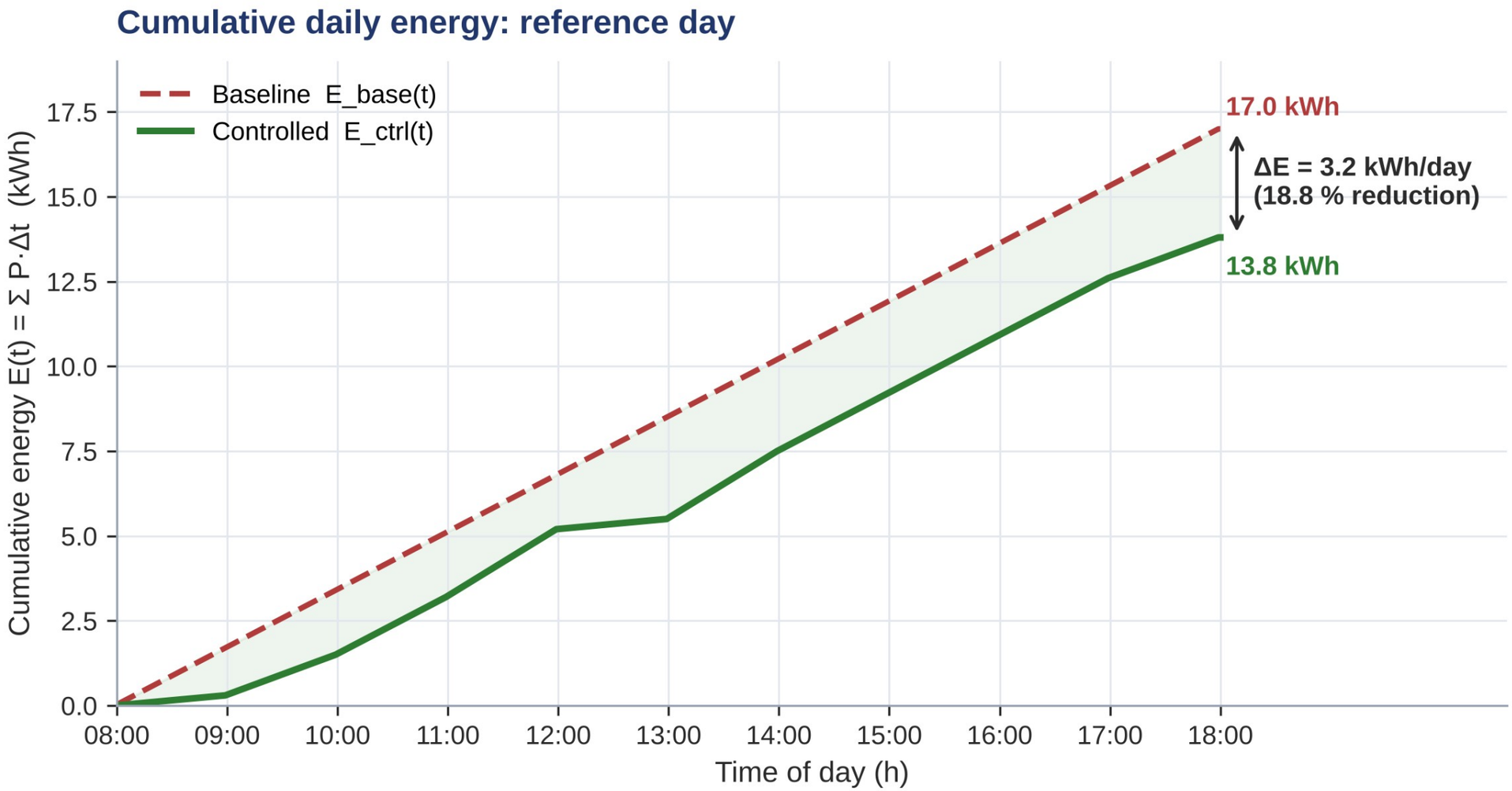
# Simulation 1: Occupancy Drives Power




*MODELLED SIMULATION · model v1 · baseline 1.70 kW, 08:00–18:00*

***Power follows people: 0.30 kW when empty, 2.00 kW when crowded, 1.70 kW flat under the schedule.***

# Simulation 2: Cumulative Daily Energy

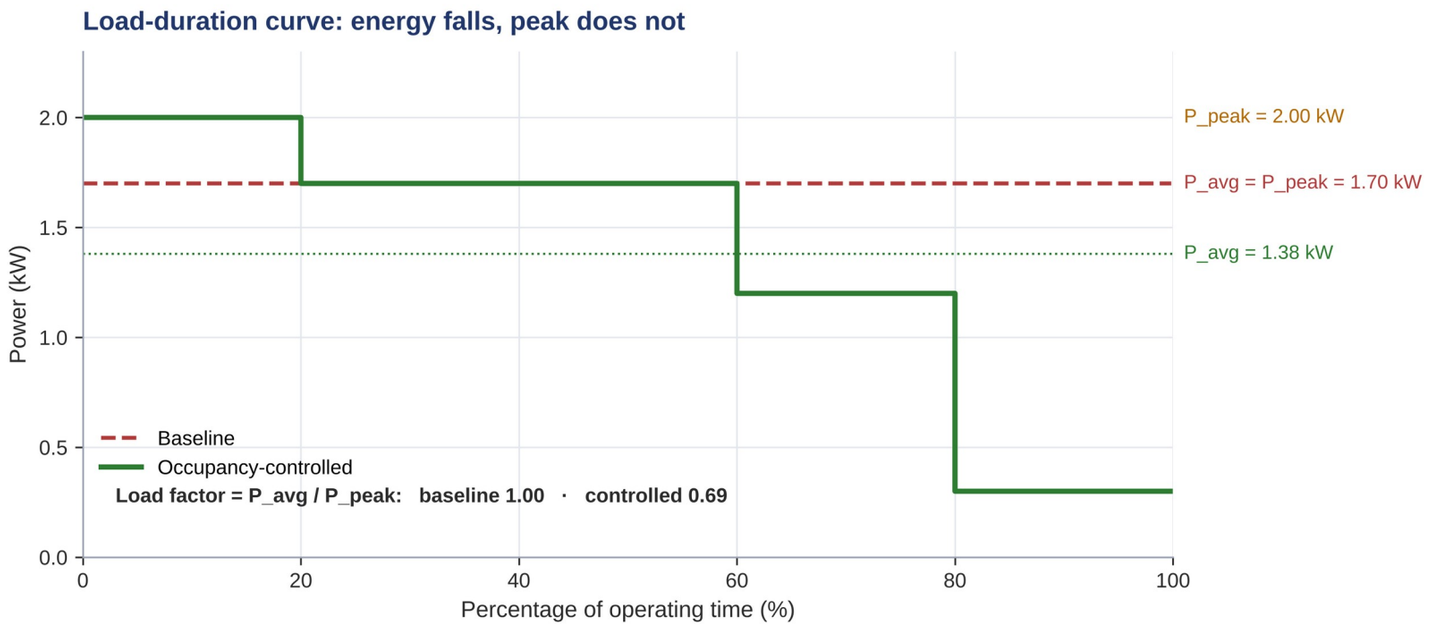



*MODELLED SIMULATION · model v1 · baseline 1.70 kW, 08:00–18:00*

***17.0 → 13.8 kWh/day: 3.2 kWh avoided (18.8 %) on the reference day.***

MODELLED

# Simulation 3: Load-Duration Curve & Load Factor




| | Baseline | Controlled |
|---|---|---|
| Average power | 1.70 kW | 1.38 kW |
| Peak power | 1.70 kW | **2.00 kW** |
| Load factor | 1.00 | 0.69 |

*MODELLED SIMULATION · model v1 · baseline 1.70 kW, 08:00–18:00*

***Energy falls 18.8 %; peak rises to 2.00 kW because a crowded room needs more cooling. We optimise energy, not peak.***

# Results Scorecard

**17.0 → 13.8 kWh/day**
baseline → controlled

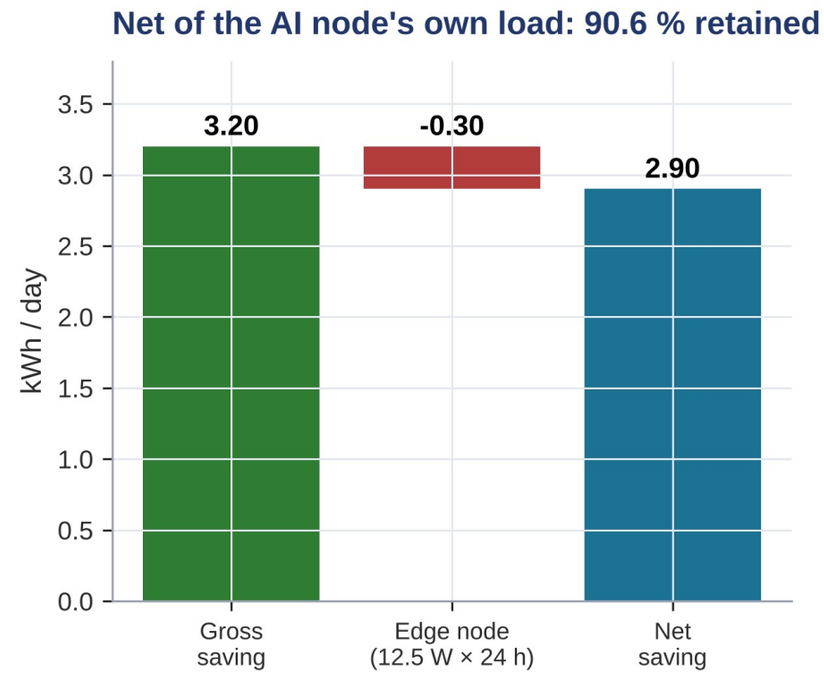
**18.8 %**
modelled energy reduction

**29.4 %**
instant cut in Economy state (matches live dashboard)

**1,168 kWh/yr**
energy avoided per room

**₹9,344/yr**
illustrative saving per room

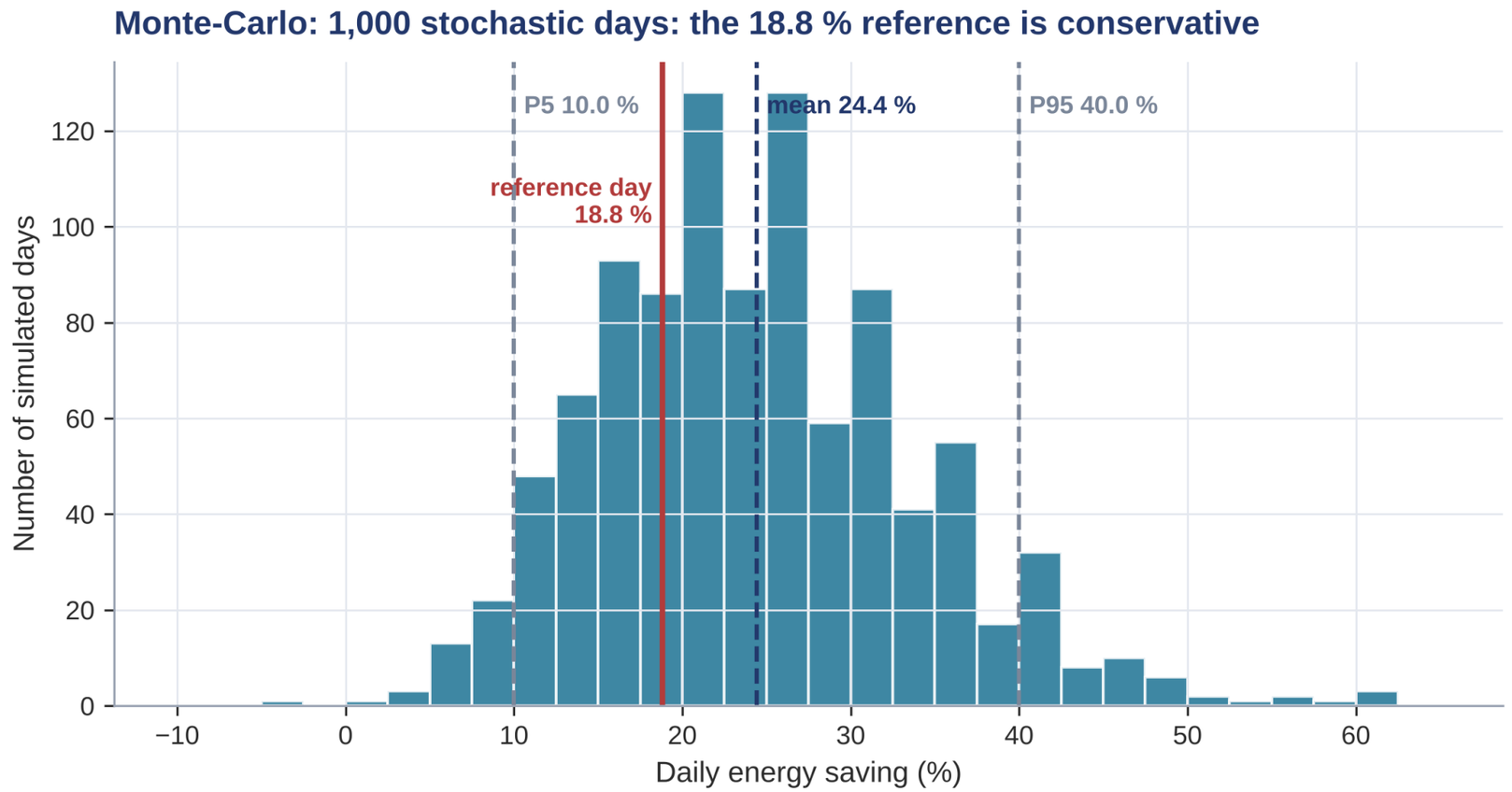
**0.83 t $CO_2$/yr**
avoided per room




*After subtracting the Jetson node's own load (12.5 W rated), 90.6 % of the saving remains.*

***Modelled energy-consumption reduction, not equipment efficiency or peak reduction.***

# Robustness: Monte-Carlo Study

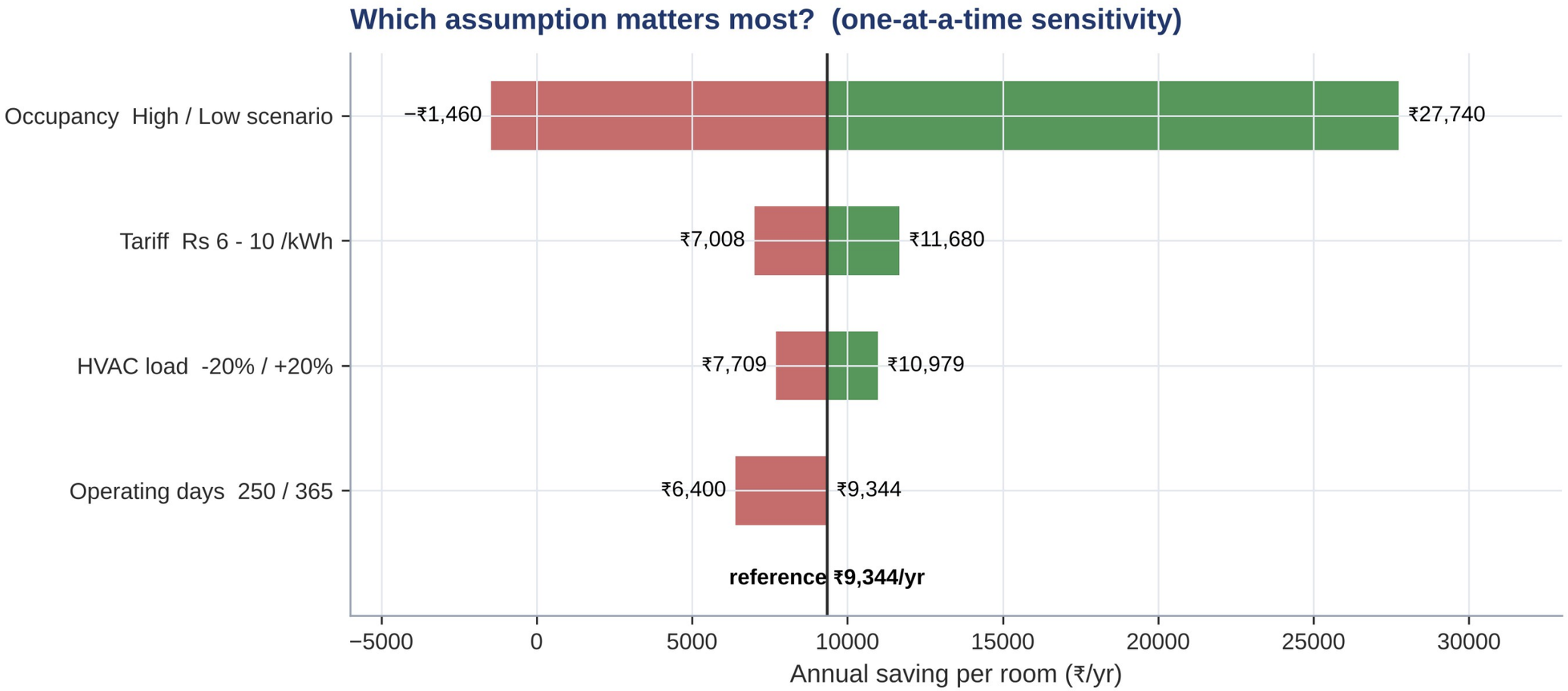



*STOCHASTIC OCCUPANCY (15 % session cancel, lognormal σ = 0.35, Poisson ad-hoc use) · seed 2026 · model v1*

**1,000** random days | mean **24.4 %** | range (P5–P95) **10–40 %**

***Across 1,000 random days the saving is positive on 99.9 % of days; our 18.8 % reference is conservative.***

# Sensitivity Analysis




*MODELLED SIMULATION · model v1 · baseline 1.70 kW, 08:00–18:00*

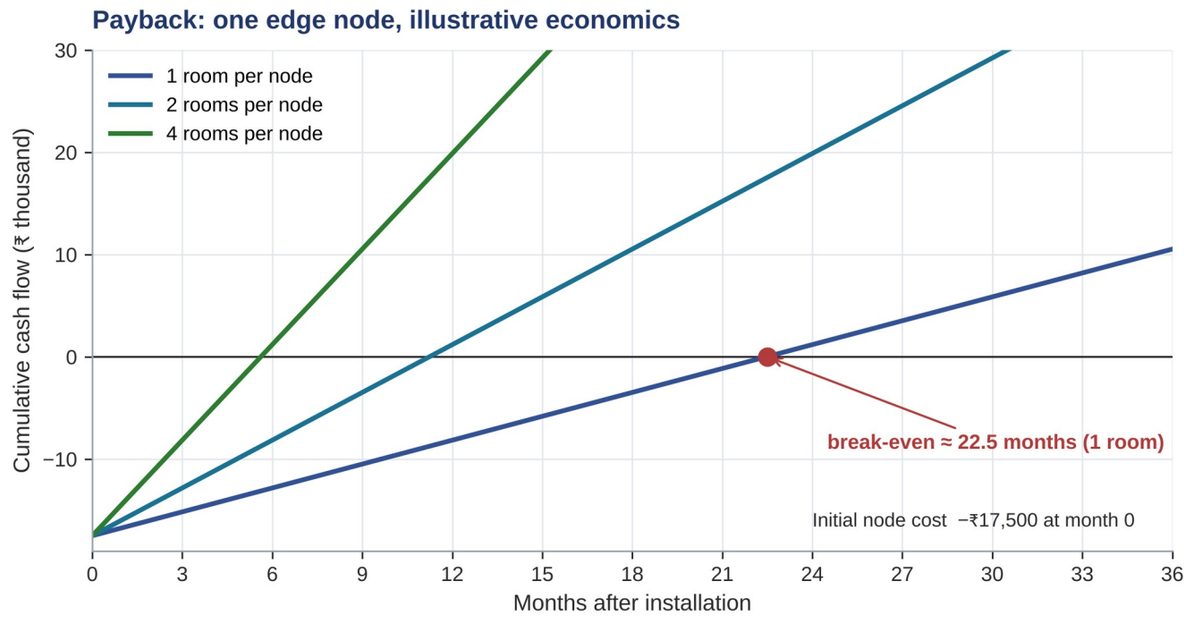
***Occupancy pattern matters most; tariff, HVAC size and operating days only scale the result.***

MODELLED (illustrative)

# Financial Plan

| Component | ₹ |
|---|---|
| Jetson Nano 4 GB Dev Kit | 13,000 |
| USB / CCTV camera* | 1,300 |
| 64 GB microSD | 700 |
| 5 V / 4 A power supply | 500 |
| Heatsink + fan | 600 |
| Wi-Fi + cabling + misc | 1,400 |
| **Total per node** | **17,500** |

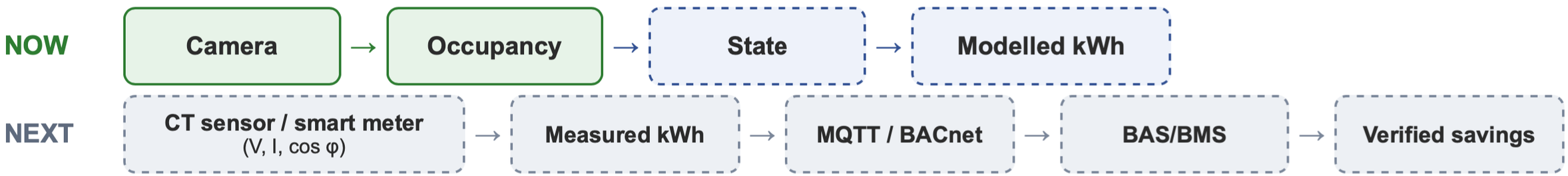
**Existing CCTV reused → camera ≈ ₹0. Software = ₹0.*




*ILLUSTRATIVE · ₹9,344 /room/yr · multi-room assumes existing CCTV per room · model v1*

***Break-even ≈ 22.5 months for one room; about 11 months if one node serves two rooms.***

# From Modelled to Measured: Roadmap & Standards




| Standard | Use | Status |
|---|---|---|
| USB / V4L2 + DeepStream | Capture + inference | IMPLEMENTED |
| ONVIF / RTSP | Existing CCTV streams | FUTURE |
| MQTT 5.0 | Occupancy telemetry | FUTURE |
| ASHRAE 135 (BACnet) | HVAC / lighting control | FUTURE |
| ISO 52120-1:2021 | Building automation energy reference | REFERENCE |

Expected meter current at 230 V: **1.45 A** empty · **9.7 A** peak → a **30 A CT** is enough

***Metering will confirm or correct the model; MQTT and BACnet are integration targets.***

# Work Completed & Plan for Final Review

## COMPLETED
**IMPLEMENTED**

✓ Jetson + DeepStream + TensorRT FP16

✓ ResNet10 live detection

✓ Occupancy count + filter

✓ Live dashboard

## COMPLETED
**MODELLED · NEW IN REVIEW 3**

✓ One simulation model

✓ Filter study

✓ Monte-Carlo

✓ Sensitivity

✓ Payback

## NEXT
**FUTURE**

→ Measure precision, recall, count error

→ Measure FPS, latency, node power

→ CT/meter kWh validation

→ MQTT demo

***Review 3 completes the simulation; the final review completes measurement.***

# Mapping with SDGs & Societal Impact

SDG 7

## Affordable & Clean Energy

Cuts avoidable energy use

SDG 9

## Industry, Innovation & Infrastructure

Adds Edge-AI to existing infra

SDG 11

## Sustainable Cities & Communities

Smarter, efficient buildings

**Privacy** · video stays on device

**Reuse** · existing CCTV, no new room sensors

**Carbon** · 0.83 t $CO_2$/room/yr, modelled

***Same cameras, new value: energy intelligence without new sensors.***

# Team Dynamics

| Paul Jey Sanjith J | Oviyan P S | Thirsna S |
| --- | --- | --- |
| • System architecture<br>• Jetson / DeepStream<br>• Simulation model | • Camera validation<br>• Testing<br>• Documentation | • Dashboard<br>• Scenarios<br>• Documentation |

**One integrated prototype with shared testing, documentation and review prep.**

# Conclusion

| | |
|---|---|
| **LIVE** | Real-time Edge-AI occupancy sensing on a Jetson Nano |
| **QUANTIFIED** | 18.8 % modelled saving, 24.4 % Monte-Carlo mean |
| **HONEST** | Energy reduced, peak not claimed |
| **SCALABLE** | Clear path to metering and BAS/BMS |

**CCTV** → **EDGE AI** → **OCCUPANCY** → **ENERGY**

## From Surveillance Data to Energy Intelligence

# Thank You !

# References


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9. Ministry of Power / BEE, default 24 °C setting for room air conditioners, effective 1 Jan 2020.
10. NVIDIA, DeepStream SDK 6.0.1 Documentation.
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